

Contrasting soil property patterns between ditch bed and neighbouring field profiles evidence the need of specific approaches when assessing water and pesticide fate in farmed landscapes

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ABSTRACT

Farm ditch networks, infrastructures designed to regulate excess water in cropped landscapes, constitute pesticide dispersal pathways or buffer zones, depending on their soil properties. Despite the key role that ditch soils play in the regulation of water and pesticide fate, their properties, especially in intermittently flooded ditches, remain poorly characterized. Therefore, our aim was to evaluate the specificity of ditch material properties to determine whether ditches require an approach that differs from that of field soils when studying water and pesticide fate in farmed landscapes. We thus analysed the variations in the pedological, herbicide sorption and flow properties of soil materials along a 2D cross-section of an intermittently flooded ditch in the Roujan catchment of southern France. We found that the upper part of the ditch bed soil profile is composed of 3 horizons that formed after the original creation of the ditch, most likely via the deposition of field-eroded particles and the accumulation of organic matter. These specific horizons have greater porosity, mostly due to their dense root systems, and contain up to 2 times more organic carbon than the neighbouring banks or field soils. Consequently, the hydraulic conductivity is greater, and the sorption of hydrophobic herbicides is up to 2 times greater in ditch bed materials than it is in soils located farther away from the ditch surface. Moreover, significant macropore flow was evidenced in both profiles but with different contribution to the global flow. The contrasts in the hydrodynamic and sorption properties between both the ditch bed and banks materials likely results in significantly different water and pesticide infiltration patterns in ditches compared to crop fields. Given these differences, we recommend investigating the specific properties of ditch beds when studying and modelling water and pesticide fate in croplands.

1. Introduction

Ditches are human-made channels arranged as networks in cropped catchments. This type of infrastructure was originally designed to drain shallow groundwater in order to control the waterlogging of arable lands and/or to collect and rapidly carry overland flow towards catchment outlets in order to prevent soil erosion (Dollinger et al., 2015a; Levavasseur et al., 2016). The elevation of the water table relative to the water level in the ditch bed determines whether the ditch will exfiltrate water from the water table or infiltrate surface water towards the water table (Dages et al., 2009; Kao et al., 2001; Needelman et al., 2007b). In temperate humid lowlands, water tables are often located above the water level in ditch beds; therefore, ex-filtration fluxes prevail, creating quasi-permanent base-flow in ditches

(Bouldin et al., 2004; Debieche et al., 2006; Needelman et al., 2007b; Vaughan et al., 2008). In contrast, in catchments located in semi-arid or arid climates, water table depths are more variable, and the ephemeral flooding of ditches leads to alternation between infiltration and ex-filtration conditions (Dages et al., 2009; Marofi, 1999). In both cases, ditches are zones of concentrated exchange between surface water and groundwater featuring two- or three-dimensional water flow patterns (e.g., Harvey and Bencala, 1993; Dages et al., 2009). Moreover, because surface runoff and drainage fluxes collected by ditches are potentially loaded with pesticides (Louchart et al., 2001; Tang et al., 2012; Voltz and Louchart, 2001), ditches may constitute contamination pathways that can rapidly carry pesticides towards both surface water and groundwater bodies. However, ditches have previously been reported to be efficient buffer zones for pesticide-based water pollution

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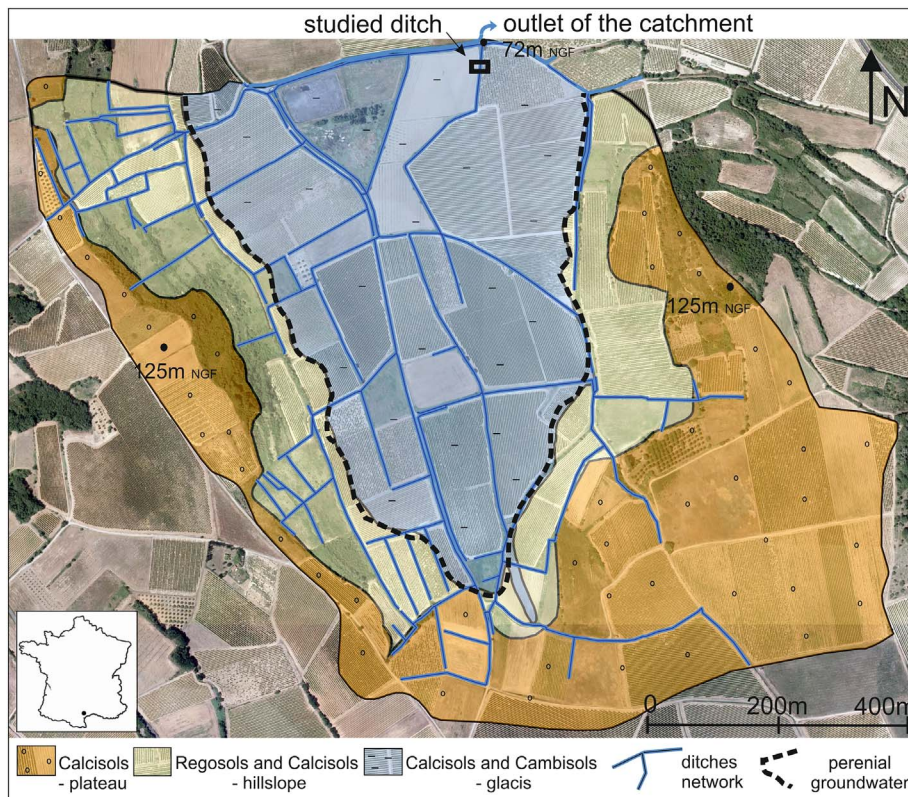


Fig. 1. The ditch network over the Roujan catchment in relation with the soils and the perennial groundwater.

(Dollinger et al., 2015a; Herzon and Helenius, 2008; Needelman et al., 2007a). The high pesticide retention capacities reported for ditches are mainly due to the sorption processes occurring on the various materials in ditch beds (Dollinger et al., 2015a; Stehle et al., 2011), especially ditch soils (Dollinger et al., 2016).

Therefore, ditches, and more specifically ditch soils, constitute key regulators of water and contaminant fluxes in cropped catchments whenever ditches are present in a landscape. This regulation role relies on several physicochemical and structural properties of the ditch materials that influence their hydraulic conductivity, their preferential flow patterns, and the sorption of contaminants, such as pesticides. However, ditch materials may differ strongly from those in neighbouring field soils for several reasons. First, because ditches are excavations, their topsoil forms from the subsoil horizons of the field soil, as well as from the deposition of field-eroded sediments and the accumulation of organic matter residue (e.g., Vaughan et al., 2008). Therefore, the topsoils of ditch beds are generally enriched in organic matter and have different textures from those of the field soil horizons located at the same depth. Second, in intensively farmed catchments, ditches often remain the only non-cropped elements in the landscape and consequently represent biodiversity hotspots (Dollinger et al., 2015a; Herzon and Helenius, 2008). Accordingly, ditches host a wide diversity of plants, rodents, insects and invertebrate species, which can potentially create soil macroporosity (Herzon and Helenius, 2008; Needelman et al., 2007b; Vaughan et al., 2008) and, in turn, produce preferential flow patterns (Dages et al., 2015). Third, ditches are subject to specific management practices that may substantially impact their physical and chemical properties. The effects of management practices on ditch characteristic are especially apparent in the case of intermittently flooded ditches as those observed by Levavasseur et al. (2014) in southern France, in which mowing, burning and chemical weeding occur at least once a year and dredging occurs once every 10 years. Some of these management operations may favour the development of

an organic horizon, while dredging involves the removal of the 15- to 30-cm-thick organic matter-rich topsoil layer. Finally, ditch soils are affected by water flow patterns that are highly different from those affecting nearby field soils. Because these flows are more intense and occur in two dimensions, over time, they can produce 2D heterogeneous hydraulic properties similar to those seen in soils surrounding buried drains (e.g., Frison et al., 2009).

Despite the key role that ditch soils play in the regulation of water and contaminant fluxes, their morphologies, as well as their properties influencing water and pesticide transfer, have rarely been studied. To our knowledge, thus far, only the morphology of lowland drainage ditches in Maryland, USA, has been studied. There, Vaughan et al. (2008) and Needelman et al. (2007b) described the occurrence of ditch bed organic horizon sequences that overlie the A, BC and C horizons with weak to moderate structuration. They identified a range of pedogenetic processes involved in the formation of ditch soils, such as organic matter humification, structure formation and bioturbation. However, they did not describe the soil properties influencing the magnitude of the water and contaminant exchanges occurring between surface water and groundwater nor did they compare these properties to those of the surrounding field soils. A different sampling scheme or assessment of water or pesticide transfer parameters may however be required for ditches as compared to crop fields if their hydraulic and physicochemical properties appear to be different and contrasted.

Therefore, the main objective of this study was to analyse the 2D heterogeneity of ditch bed and bank soil materials in an intermittently flooded ditch in order to evaluate whether ditches require a different approach from that of field soils when studying and modelling water and pesticide transfer in farmed landscapes. With this aim, the variations in the pedological, sorption and flow properties of soil materials were characterized along a 2D cross-section of a representative ditch in the Roujan catchment in southern France.

Table 1
The spatial variability of ditches network over the catchment.

Topographic position	Dominant soil types (WRB 2014)	Soil depth	Surface area	Total length of ditch in the network
		m	ha	km
Plateau Hillslope	Calcisols	0.4–2	36	1.49
	Regosols and calcisols	0.2–1.5	22	4.78
Glacis	Calcisols and cambisols	1–4	33	5.48
<i>Catchment</i>			91	11.75

2. Materials and methods

2.1. Study site

The studied ditch is located near the outlet of the Roujan catchment (Hérault, France). This 91 ha catchment (Fig. 1 and Table 1) is cultivated mainly by vineyards and a dense network of ditches, 11 km total length, was implemented between the vine fields. Except on the plateau, the soils are directly developed over the Miocene loose sandstone and are organized along a toposequence. The soils depth increase and soil texture evolve consistently with the colluvial accumulations of clay and gravels in the glacis. Nearby the study site the soil is classified as a gleyic cambisol (IUSS Working Group WRB, 2014). A perennial groundwater has developed on the bottom part of the catchment (Fig. 1) and > 5 km of ditches (47% of the total length of ditches) drain this area.

The catchment is subjected to semi-arid Mediterranean climate characterized by scarce high-intensity rainfall events (Andrieux et al., 1993). This specific precipitation pattern results in the periodic flooding of ditches and the rapid fluctuation of the shallow water table in the bottom part of the catchment (Dages et al., 2009). The high reactivity of the water table leads to the alternation of downward and upward fluxes in ditch beds during storm events (Dages et al., 2009).

The studied ditch is chosen near the catchment outlet in order to: i) represent the typical functions of ditch in a perennial groundwater environment (Fig. 1) and ii) be representative of the soil type and ditch characteristics combination that prevail in the 33 ha of the bottom part of the catchment.

2.2. Experimental design

2.2.1. Characterization of soil properties and core sampling along the cross section

For characterizing and sampling soil heterogeneity of the ditch soil and its vicinity, a 1.50-m-wide, 1.50-m-deep trench was excavated across the ditch in February 2014. The studied ditch is densely vegetated, and roots are present along the entire soil profile to a depth of 1.5 m (Figs. 2A and 3A).

A series of morphological parameters, including texture, structure, colour, stone and root abundances, were observed in the field. Soil horizons were determined based on these observations. Bulk densities (ρ_b) were measured by core sampling with 100-cm³ cylinders (Fig. 2B), using 6 replicates per horizon. The ρ_b was determined as the ratio between the dry soil mass and the total core sampling volume. Samples of over 500 g were collected from each horizon for further laboratory characterization. Particle size distribution, pH, cation exchange capacity (CEC), organic carbon content (OC), and calcium carbonate (CaCO₃) content were measured at the INRA-ARRAS Laboratory (France) (see Table 2).

Four undisturbed soil cores were sampled from each horizon (Fig. 2B) except in the Ap horizon of the bank profile that has no

counterpart in the ditch soil profile. These cores were collected by gently pushing stainless-steel cylinders with internal diameters of 15 cm and heights of 20 cm in the soil until the soil surface was approximately 5 cm from the top of the cylinder (Fig. 2C). The soil around the cylinders was then excavated to facilitate the undisturbed extraction of the monoliths. To characterize the anisotropy of downward vs. lateral water and solute flow, a series of monoliths was sampled vertically and a second series was sampled horizontally (Fig. 3B). Due to the length of the sampled cores, the core sampled in the first horizon below the ditch also included the top of the second horizon; because the third horizon below the ditch was too narrow, it could not be sampled. After extraction, cores were stored at 4 °C until undergoing tracer experiments.

2.2.2. Tracer displacement experiments

Tracer displacement experiments were performed on soil cores sampled vertically and horizontally in the ditch bed and bank profiles (Fig. 3B) in order to characterize the water flow patterns of these materials. Because bromide is only present at trace concentrations in the environment and rarely sorbs to soil particles (Flury and Wai, 2003), it was selected as a conservative tracer of water flow for these displacement experiments.

Stainless-steel grids with 6-mm-diameter holes were sealed at the bottom of the columns to prevent soil loss occurring during the infiltration experiments without disturbing the water flux in the columns. Prior to tracer injections, the columns were gradually saturated via capillarity for 48 h to prevent the trapping of gas bubbles in soil pores.

The tracer solutions used for the displacement experiments contained 800 mg l⁻¹ of bromide (Br⁻). At the beginning of the displacement experiments, the saturated columns were manually ponded with a 30-mm water height of the tracer solution. This water height was chosen to mimic the infiltration conditions in the Roujan catchment during intermittent flooding and corresponds to the water level commonly monitored in ditches during flood events with a 1-month return period. The water height was kept constant during the infiltration by adjusting the supply of the tracer solution. A total of 85 mm of solution was supplied to the columns. Among the 16 columns, the pore volumes ranged from 63 to 82 mm. Therefore, the volume of tracer solution supplied during the displacement experiment was always higher than the pore volume of the columns. When the solution supply stopped, the decrease in water head was monitored during the remaining period of ponded water infiltration. Just after all the ponded solution had infiltrated, the columns were ponded again with a constant head of 30 mm, and the columns were flushed with 85 mm of tap water, following the same procedure. During the infiltration and flushing periods, 50 ml fractions of the percolates were collected in glass containers at the outlets of the columns. The sampling frequency varied from 15 s to about 6 min, depending on the columns drainage fluxes. The outlet flowrates were monitored using the timing of sample collection and their precise weights. The inlet flowrates were monitored by weighing the injection tank at 1-s intervals.

The concentrations of bromide in the percolate samples were measured using an ion-specific electrode (Hanna Instruments, HI4002, Lingolsheim). These concentration values were cross-validated with ion chromatography measurements of randomly selected samples. A good fit was found between the ion-specific electrode and the ion chromatograph results (data not shown). The electrical conductivity and pH were also measured in the percolate samples.

2.2.3. Dye tracing of the active macroporosity

Following the displacement experiments, dye tracing was performed on the columns to visualize and quantify the active macroporosity. The dye tracing experiments were also used to visualize the presence or absence of sidewall flow. The displacement experiments were validated when sidewall flow was absent or weak and discontinuous along the sides of the columns. In contrast, when continuous sidewall flow was



Fig. 2. A: studied ditch near the outlet of the Roujan catchment. B: core sampling for bulk density measurement. C: core sampling for water and bromide displacement experiments.

detected along the sides of the columns, the corresponding displacement experiments were dismissed. A total of 8 columns, containing one sample per horizon and one sample per direction (vertical/lateral) were validated based on dye staining experiments.

The infiltration conditions of dye tracing were identical to those of the displacement experiments, as the fraction of active porosity mobilized for percolation in structured soils likely varies with initial moisture and water head conditions (Sammartino et al., 2015;

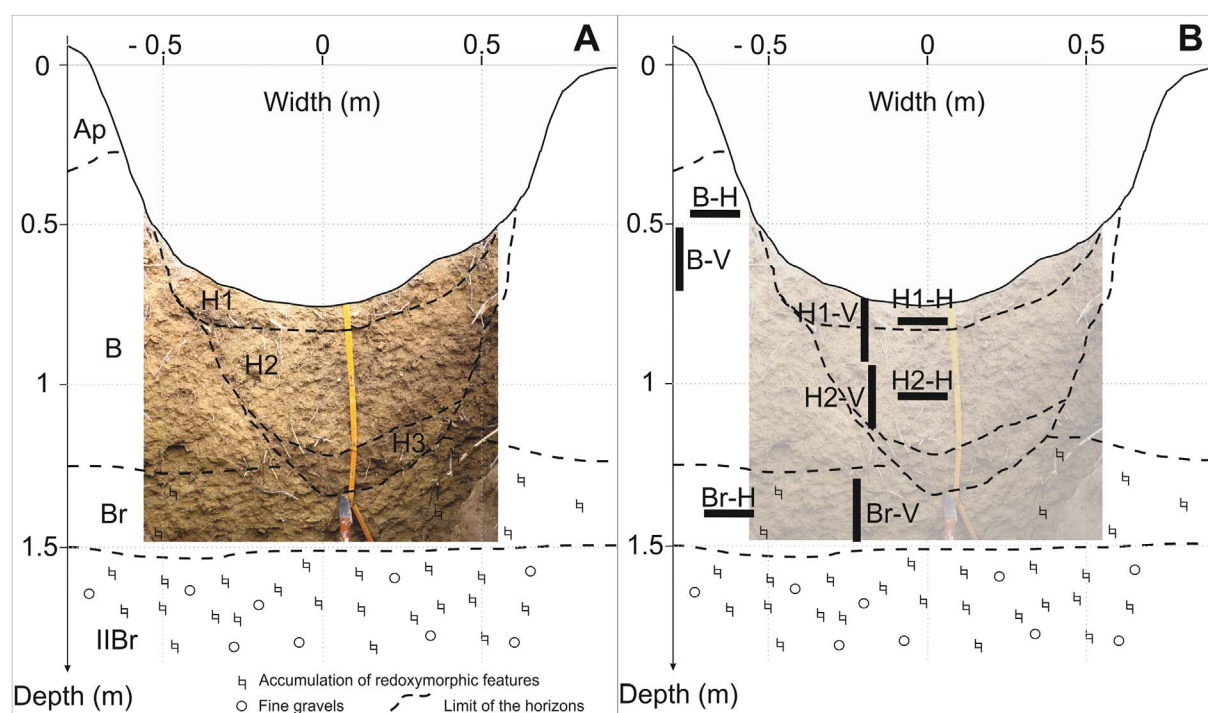


Fig. 3. Morphology of the ditch cross-section soil profile. A) Description of the soil profile, B) core sampling scheme. The black lines represent core sampling locations within the soil profile. V and H represent the cores sampling axis direction being, respectively, vertical and horizontal.

Table 2
Physico-chemical properties of ditch-bed and banks soils.

Horizon	Depth from field topsoil m	Structure	Sand %	Silt %	Clay %	OC %	CEC cmol kg ⁻¹	pH	CaCO ₃ g kg ⁻¹	ρ _b g cm ⁻³
B	0.4–1.30	Polyhedral subangular blocky – hydromorphic features	7.7	57.2	35.1	0.96	14.2	8.71	247.0	1.36 ± 0.01
H1	0.75–0.82	Stratified	35.9	39.1	25.0	1.58	12.5	8.44	150.0	1.25 ± 0.06
H2	0.82–1.15	Granular	40.3	33.1	26.6	1.56	12.1	8.54	160.0	1.26 ± 0.04
H3	1.15–1.35	Stratified and subangular blocky – hydromorphic features	10.4	56.7	32.9	1.17	14.0	8.59	248.0	1.21 ± 0.05
Br	1.30–1.50	Polyhedral blocky – hydromorphic and redoximorphic features	4.7	58.5	36.8	0.73	14.4	8.63	285.0	1.48 ± 0.03

Vanderborcht et al., 2002). The columns were thus saturated again via capillarity for 48 h; then, 57 mm of the fluorescent dye sulforhodamine B at a concentration 1 g l⁻¹ was percolated through the columns with a constant water head of 30 mm. At a concentration of 1 g l⁻¹, the sorption sites of sulforhodamine B on soils in contact from all horizons were saturated (Dollinger et al., 2017), which guaranteed homogeneous staining among the columns. After percolation, the columns were sliced into cross-sections approximately 2 cm in height. A marker was placed on the sides of the slices to determine the orientation and superposition of the 7 slices within a given column. The slices were then imaged in a dark chamber with homogeneous LED lighting (3800 K) using a digital camera that was equipped with a 28-mm lens and was positioned 65 cm above the slice. The image resolution was 300 dpi, which corresponds to a pixel size of 71 μm. The illumination and hue saturation of the raw images were corrected using Nikon Capture NX2 software based on the grey and colour scales positioned next to the column slices during imaging. The RGB channels were split, and the colour thresholds were adjusted in each of the channels. The minimum/maximum thresholds applied to all images were 9/255, 118/253 and 4/255 for the R, G and B channels, respectively. The RGB channels were then merged, and the image was binarized. Both bright and dark isolated pixels were removed using the ‘Noise’ function of the ImageJ software (Schneider et al., 2012), with a radius of 10 pixels, for white and black pixels, successively. The respective areas of both bright and dark pixels relative to the total area of the column cross-section were then calculated with ImageJ. The dark areas correspond to the stained areas on the cross-sections of the columns.

The volume of macroporosity mobilized during percolation relative to the total porosity (ω) was estimated from the dye coverage area. As dye diffusion in the matrix is limited, due to its short infiltration time, ω_i was calculated for each column cross-section (i) by multiplying the average dye coverage area per column slice (i.e., top and bottom coverage) by the slice height and dividing it by the total porosity (i.e., the total volume of soil in the slice multiplied by the soil porosity). The average ω per column was also calculated as a geometric mean of the respective ω_i values of the 7 column slices (i).

2.2.4. Inverse modelling of transport properties

2.2.4.1. Water flow and transport equations.

Inverse modelling was performed with the HYDRUS-1D (Šimůnek et al., 2003; Šimůnek and van Genuchten, 2008) that solves the Richards and convection-dispersion equations. Four modelling approaches were compared in the first place: single porosity, dual porosity, dual porosity + mobile-immobile (DP + MIM) and dual permeability (see Šimůnek et al., 2003 for a detailed description of these approaches). Only the dual-permeability model provided satisfactory fits of the tracer displacement experiments for most of the columns and is thereby considered in this paper. The column H2-H was the only exception for which the model DP + MIM was better adapted than the dual permeability model. DP + MIM was used to simulate the bromide breakthrough curve of H2-H but is not described in this paper (for the description of the model please refer to Šimůnek et al., 2003). Equations of the dual permeability model are briefly reviewed below. For more details on equilibrium and nonequilibrium formalisms, see for

example Šimůnek and van Genuchten (2008) or Köhne et al. (2009).

The dual permeability model assumes that flow and solute transport occur within and between two distinct compartments, namely the macropore compartment, consisting in inter-aggregate or fracture porosities, and the micropore or matrix compartment, consisting in intra-aggregate porosity. The water flow equations in the macroporal and matrix compartments are assumed similar by HYDRUS 1D and given by:

$$\frac{\partial \theta_f(h_f)}{\partial t} = \frac{\partial}{\partial z} \left[K_f(h_f) \left(\frac{\partial h_f}{\partial z} + 1 \right) \right] - S_f(h_f) - \frac{\Gamma_w}{\omega} \quad (1a)$$

$$\frac{\partial \theta_s(h_s)}{\partial t} = \frac{\partial}{\partial z} \left[K_s(h_s) \left(\frac{\partial h_s}{\partial z} + 1 \right) \right] - S_s(h_s) - \frac{\Gamma_w}{1 - \omega} \quad (1b)$$

where subscript *f* and *s* respectively refers to the fast macroporal compartment and the slow matrix compartment, *θ* is the water content [L³·L⁻³], *h* is the pressure head [L], *K(h)* is the unsaturated hydraulic conductivity function, *S* is a sink or source term [T⁻¹], ω is the ratio of the macroporal volume of fast to the total poral volume of the soil (dimensionless) and Γ_w is the transfer rate between the two compartments [T⁻¹]. The water retention curve *θ(h)* and the unsaturated hydraulic function *K(h)* are defined for both compartments using the van Genuchten model (Van Genuchten, 1980). *K(h)* is described as the product of the relative hydraulic conductivity function *K_r* (dimensionless) and the saturated hydraulic conductivity *K_s* [L·T⁻¹].

The transport equations associated with the dual-permeability formulation for water flow are based on the classical convection-dispersion equation for both the fast macroporal compartment and the slow matrix compartment with an exchange term between the two compartments:

$$\frac{\partial \theta_f c_f}{\partial t} + \rho \frac{\partial s_f}{\partial t} = \frac{\partial}{\partial z} \left(\theta_f D_f \frac{\partial c_f}{\partial z} \right) - \frac{\partial q_f c_f}{\partial z} - \phi_f - \frac{\Gamma_s}{\omega} \quad (2a)$$

$$\frac{\partial \theta_s c_s}{\partial t} + \rho \frac{\partial s_s}{\partial t} = \frac{\partial}{\partial z} \left(\theta_s D_s \frac{\partial c_s}{\partial z} \right) - \frac{\partial q_s c_s}{\partial z} - \phi_s + \frac{\Gamma_s}{1 - \omega} \quad (2b)$$

$$\Gamma_s = \omega_{dp}(1 - \omega)\theta_s(c_f - c_s) + \Gamma_w c^* \quad (2c)$$

where *c* is the solute concentration [M L⁻³], *s* is the sorbed solute concentration [M M⁻¹], ρ is the bulk density [M L⁻³], *D* is the dispersion coefficient accounting for both molecular diffusion and hydrodynamic dispersion [L² T⁻¹], *q* is the Darcian flux [L T⁻¹], φ is a sink-source term [M L⁻³ T⁻¹], Γ_s is the mass transfer term for solute between the macroporal and the matrix compartments [M L⁻³ T⁻¹] and *c*^{*} is equal to c_f for Γ_w > 0 c_m for Γ_w < 0.

2.2.4.2. Inverse modelling design.

The 0.15 m-soil profiles were densely discretized with 101 nodes to facilitate numerical convergence. An initial hydrostatic equilibrium with a zero-pressure at the top of the soil columns was considered. A variable head was imposed at the upper boundary condition. It was fixed at the ponding head value (3 cm) during the injection and rinsing phases and varied between both phases to correspond to the ponding height decreases monitored during the experiments (see Section 2.2.2). The eight following parameters were fitted against the cumulative water outflows heights and bromide concentrations at the

outlet of the soil column: $\theta_s, \theta_{sf}, K_{s_s}, K_{s_f}, \omega, Disp_s, Disp_f, \omega_{dp}$ with θ_s and θ_r respectively the saturated and residual soil water content and $Disp$, the dispersion coefficient [L]. To avoid local minimum, the stability of the fitted parameter set estimated was evaluated using different sets of initial parameters, including the estimated sets themselves. The other hydrodynamic parameters were set according to the textural composition and bulk density of the soils using Rosetta (Schaap et al., 2001), except θ_{rf} that was set to zero. Note however that since the soil column remained saturated during the whole experiment, the van Genuchten parameters α , n and l were not sensitive. The Bromide diffusion coefficient was fixed to $1.67 \cdot 10^{-9} \text{ m}^2 \text{ s}^{-1}$. K_{sat} was not adjusted but calculated from the experimental outflow data using Darcy's law. Θ_s was also not adjusted but calculated from the bulk density data using a pedotransfer function.

2.2.5. Sorption properties of selected herbicides

Two herbicides, diuron and glyphosate, were selected to assess the heterogeneity of sorption properties along the profile of the ditch cross-section. Diuron was extensively used on the Roujan catchment for weed control in vineyards. After it was banned from the list of allowed active molecules in France in 2008, it was replaced by the broad-spectrum herbicide glyphosate. Both herbicides were still measured in the water column of the ditch at the outlet of the catchment in 2016. Glyphosate and diuron exhibit very different physicochemical properties (Table 3), which may lead to contrasting sorptive patterns along the soil profiles.

The adsorption parameters were assessed according to the procedure described in Dollinger et al. (2016), which was adapted from the OECD Guideline 106 (OECD, 2000). Briefly, the soils were air-dried to a target humidity of 10% then sieved to a size of 2 mm. 10 ml of the ^{14}C -labelled pesticide solution, with concentrations ranging from 5 to $1000 \mu\text{g l}^{-1}$, were equilibrated with 1 and 2 g of dry soil in glass centrifuge tubes for glyphosate and diuron adsorption experiments, respectively. The tubes were shaken for 24 h, and the radioactivity in the supernatant was measured after centrifugation. Pesticide concentrations in soils were assessed by mass balance between initial and equilibrium concentrations. Both linear (Eq. 3) and Freundlich models (Eq. 4) were fitted to the experimental data.

$$C_s = K_d C_w \quad (3)$$

$$C_s = K_f C_w^n \quad (4)$$

$$H = \frac{n_{des}}{n_{ads}} \quad (5)$$

where C_s is the amount of sorbed pesticides in the soil at equilibrium ($\mu\text{g kg}^{-1}$), C_w is the equilibrium concentration in the supernatant ($\mu\text{g l}^{-1}$), K_d is the linear sorption coefficient (l kg^{-1}), K_f ($\mu\text{g}^{(1-n)} \text{ l}^n \text{ kg}^{-1}$) and n are the Freundlich coefficients and H is the apparent hysteresis index with n the non-linearity parameter of the Freundlich model and subscripts ads and des standing for adsorption and desorption isotherms, respectively.

The detailed procedure for the determination of herbicide desorption parameters can be found in Dollinger et al. (2016). Briefly, after 24 h of equilibration with a $100 \mu\text{g l}^{-1}$ pesticide solution, the activity in the supernatant was measured and the residual supernatant was removed. An equivalent volume of fresh electrolyte was added, and the

tubes were shaken again for 24 h. Five successive desorption steps of 24 h each were then performed. The amount of pesticides sorbed to soils at each step was calculated by mass balance based on radioactivity counting, and experimental data were fitted to Freundlich isotherms (Eq. 5). The hysteresis between adsorption and the corresponding desorption isotherms was represented by the H parameter (Eq. 3), which was calculated as proposed by Barriuso et al. (1994). Sorption is considered to be hysteretic when $H < 0.7$; the lower the value of H is, the more irreversible the sorption is.

3. Results and discussion

3.1. Morphology of the ditch bed and bank profiles

Based on field morphological descriptions, two different soil profiles were distinguished along the cross-section: (i) the bank profile, which is composed of 4 horizons, and (ii) the ditch bed profile, which is composed of 5 horizons (Table 2 and Fig. 3).

The bank soil profile corresponds to the soil pit observed in the vicinity of the ditch by Andrieux et al. (1993). According to the World Reference Base (IUSS Working Group WRB, 2014), this soil is a tilled gleyic Cambisol (colluvic, clayic). The structure of the first horizon (Ap, which extends from the surface to a depth of 0.4 m) is affected by tillage and deep ploughing operations (Coulouma et al., 2006). The upper cambic horizons B and Br (described in Table 1) are developed above another deep cambic horizon IIBr (Fig. 3A) that feature both a high clay content and high bulk density values. However, more hydromorphic features and denser root systems are observed closer to the ditch bank surface than they are in the bank soil profile, which is located further away. The ditch bed soil profile corresponds to a succession of 3 ditch-specific horizons (H1, H2, and H3) and the Br and IIBr horizons, which are shared with the bank profile. The H1, H2, and H3 horizons are significantly different from the other horizons. H1 and H2, which are enriched in sand and have platy structures, are different from the bank horizon B, which is siltier and is dominated by a subangular blocky structure. The third horizon, H3, is similar to the bank horizon B in terms of texture but features a stratified structure that differs from that of the bank horizons. These differences indicate that the H1, H2, and H3 horizons were formed by the deposition of field-eroded particles during successive flood events (e.g., Vaughan et al., 2008) subsequent to the creation of the ditch (Dages et al., 2015). The contours of the 3 horizons specific to the ditch bed (H1, H2 and H3; Fig. 3A) thus likely delimit the section of the original ditch. The shape of the horizons is probably due to the regular management of the ditch, including dredging operations. At the location where the profiles were observed, the original ditch only slightly incises the Br horizon that prevailed prior to the creation of the ditch.

The B and Br horizons have very similar physicochemical properties (Table 2), although horizon B has a slightly greater organic carbon content and a lower bulk density. However, the porosities of horizons B and Br are larger in the vicinity of the ditch surface due to the higher density of the ditch vegetation root system (Fig. 2). The upper two ditch bed soil horizons (H1 and H2) contain 1.5 to 2 times more organic carbon than horizon B, which is consistent with the presence of vegetation and higher water contents during the year. Moreover, the bulk densities of the specific ditch bed horizons are significantly lower than those of the other horizons, which is in accordance with their textures, organic matter contents, and dense root channels network. Therefore, the overall porosity of the ditch bed soil profile is higher than that of the bank profile.

In summary, the ditch bed profile and the bank profile have contrasting textural, chemical and structural properties. Moreover, the vertical gradient of the analysed soil properties across the ditch bed horizons is sharper than that across the bank horizons. Both lateral and vertical gradients of soil properties, such as organic matter content or bulk density, are present within the limited spatial area of one square

Table 3
Physico-chemical properties of the studied pesticides.

Properties		Glyphosate	Diuron
Formula		$\text{C}_3\text{H}_8\text{NO}_5\text{P}$	$\text{C}_9\text{H}_{10}\text{Cl}_2\text{N}_2\text{O}$
Molecular mass	g mol^{-1}	169.1	233.1
Aqueous solubility at 20 °C	g l^{-1}	10.5 to 12.0	0.42
Log Kow at pH 7		− 4.1 to − 3.2	2.7
$\text{pK}_{a1} - \text{pK}_{a2} - \text{pK}_{a3}$		2.2–5.5–10.2	13.2

From ANSES, 2017, FOOTPRINT, 2015, ChemID, 2017 and chemicalize.org 2017.

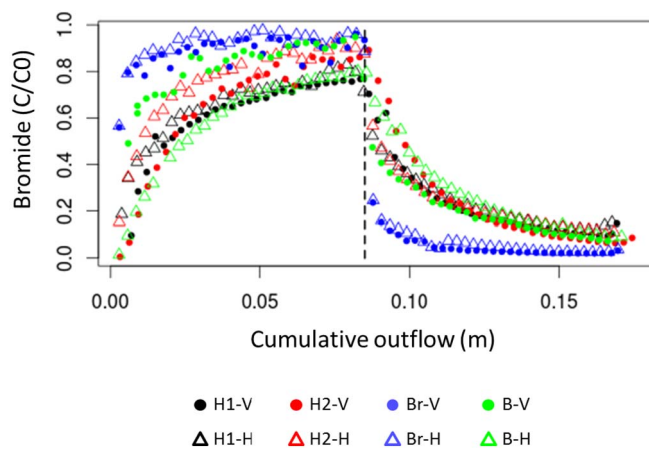


Fig. 4. Bromide breakthrough curves. The black dashed lines represent the shift between contaminated and clear water injection.

metre between the ditch and the bank. It is therefore expected that flow and sorption properties differ between the ditch and bank soils.

3.2. Heterogeneity and anisotropy of water pathways and associated soil pore structure

The results of the displacement experiments and dye staining of the active macroporosity (ω_{dye}) allowed us to compare the hydraulic conductivity and preferential flow patterns of the different horizons and sampling axes in the two soil profiles (Fig. 4, Table 4) and to relate these flow patterns to the macroporosity patterns (Figs. 5 and 6). The inverse modelling procedure provides a complementary estimation of the flow mechanisms and soil hydrodynamic properties (Table 4). The main interest of the modelling results is the opportunity to estimate the contribution of the fast flow to the global outflow.

The horizontal and vertical saturated hydraulic conductivity values (K_{sat}) at the column scale calculated from the percolation flux data range from very large ($1.7 \cdot 10^{-4} \text{ m s}^{-1}$ for H2-V) to rather small ($6.9 \cdot 10^{-6} \text{ m s}^{-1}$ for Br-H) values (Table 4). Regardless of the horizon, no systematic differences were observed in the measured K_{sat} values between the two sampling axes. With the exception of the second horizon in the ditch bed (H2), the anisotropy of the hydraulic conductivity was small in all samples. Therefore, a mean saturated hydraulic conductivity value was calculated for each horizon, and these values are reported in Table 4. The mean saturated hydraulic conductivity of the B horizon is slightly smaller than those of the H1 and the H2 horizons,

despite important differences in their textures, organic matter contents and structures, as observed in Section 3.1. Horizon H2 is the most conductive horizon due to the large value of its observed vertical K_{sat} , which may be caused by specific local macropore features, such as the snail shells observed in this horizon. The Br horizon is 4 to 15 times less conductive than the other horizons. Accordingly, as generally observed in structured soils, both ditch bed and bank soil profiles exhibit decreasing soil hydraulic conductivity with depth (e.g., Sammartino et al., 2015; Udawatta and Anderson, 2008), but the upper horizons of the ditch bed profile have higher permeability values than those of the bank profile.

The dye tracing experiments reveal information about the active macroporosity patterns (Fig. 5) and, thus, about the heterogeneity of the soil structures between the horizons. In all columns, the dye percolated across the column demonstrating the presence of connected macroporosity along the height of the column. However, the magnitude of this connected macroporosity varied greatly between the horizons. Roots were found to be the main source of flow paths in most horizons, as most stained areas surrounded living or decayed root channels. However, not all living or decayed root channels were stained. Roots were present throughout the entirety of both profiles, but denser networks were located near the ditch surface (Fig. 2). Consistently, on average, the active porosity was largest in the cores sampled in the upper horizons of the ditch bed profile (Figs. 3 & 5, Table 4). The B horizon exhibits a large anisotropy in ω_{dye} , yielding a very large value of approximately 20% in the horizontal direction. This anisotropy may be partly explained by the fact that the sampling location of the horizontal column is almost in the ditch sidewalls and is slightly further away for the vertical column (Fig. 3). The H1 and H2 horizons both exhibit a large active porosity, as H2 has the largest ω_{dye} values of all of the horizons. The numerous snail shells, combined with the granular structure present in H2, are likely responsible for its greater active porosity than H1. The active porosity of the Br horizon is significantly smaller than those of the other horizons. Finally, in accordance with the observed variations in saturated hydraulic conductivity, the ditch bed profile exhibits, on average, a larger active porosity than the bank profile. Indeed, although the linear correlation is not statistically significant, K_{sat} generally increases when ω_{dye} increases (Fig. 6).

Fig. 4 depicts the succession of bromide concentrations measured at the column outlets during displacement experiments. The results are expressed as the ratio of outlet to inlet concentrations. These concentration evolutions could all be satisfactorily simulated using a dual-permeability model except the H2-H for which a dual-porosity model with a mobile-immobile conceptualization of the matrix compartment was needed. For each set of parameters allowing a good reconstitution

Table 4
Hydrodynamic properties of the ditch-bed and banks soil horizons.

Columns	$^a K_{sat}$	$^a \text{Mean } K_{sat}$	$^a \omega_{dye}$	$^b \theta_{r_s}$	$^b \theta_{s_s}$	$^b K_{s_s}$	$^b \theta_{s_f}$	$^b K_{s_f}$	$^b \omega$	$^b \text{Disp}_s$	$^b \text{Disp}_f$	$^b \theta_{s_{Tot}}$	$^a \theta_s$	$^b \text{Relative contribution of preferential flow}$
	m s^{-1}	m s^{-1}	%	$\text{m}^3 \text{ m}^{-3}$	$\text{m}^3 \text{ m}^{-3}$	m s^{-1}	$\text{m}^3 \text{ m}^{-3}$	m s^{-1}	%	m	m	$\text{m}^3 \text{ m}^{-3}$	$\text{m}^3 \text{ m}^{-3}$	%
B-V	$2.34 \cdot 10^{-5}$	$2.68 \cdot 10^{-5}$	5.7 ± 3.5	0.093	0.55	$3.01 \cdot 10^{-5}$	0.150	$3.07 \cdot 10^{-4}$	10.0	0.13	0.5	0.51	0.48	53.1
B-H	$3.02 \cdot 10^{-5}$		20.6 ± 3.5	0.093	0.45	$8.95 \cdot 10^{-6}$	0.492	$7.76 \cdot 10^{-5}$	21.0	0.27	24.25	0.46	0.48	69.7
H1-V	$2.99 \cdot 10^{-5}$	$4.41 \cdot 10^{-5}$	12.0 ± 3.6	0.075	0.41	$1.35 \cdot 10^{-5}$	0.367	$1.13 \cdot 10^{-4}$	17.0	0.1	0.07	0.40	0.52	63.2
H1-H	$5.82 \cdot 10^{-5}$		14.4 ± 2.8	0.075	0.45	$2.26 \cdot 10^{-5}$	0.450	$3.79 \cdot 10^{-4}$	10.0	0.23	30	0.45	0.52	65.1
H2-V	$1.70 \cdot 10^{-4}$	$1.05 \cdot 10^{-4}$	20.8 ± 5.2	0.077	0.55	$4.91 \cdot 10^{-5}$	0.440	$5.98 \cdot 10^{-4}$	22.0	0.1	0.1	0.52	0.52	77.5
H2-H**	$3.99 \cdot 10^{-5}$		18.0 ± 3.5	0.077	0.49	$1.18 \cdot 10^{-5}$	0.430	$3.0 \cdot 10^{-4}$	10.0	0.03	25.84	0.48	0.52	73.3
Br-V	$6.90 \cdot 10^{-6}$	$7.38 \cdot 10^{-6}$	2.2 ± 2.0	0.092	0.40	$1.08 \cdot 10^{-6}$	0.300	$9.80 \cdot 10^{-5}$	6.0	0.1	0.47	0.39	0.44	85.3
Br-H	$7.86 \cdot 10^{-6}$		0.7 ± 1.1	0.092	0.40	$1.08 \cdot 10^{-6}$	0.300	$9.80 \cdot 10^{-5}$	7.0	0.1	0.47	0.39	0.44	87.2

^a Values obtained from the experimental data.

^b Values obtained from the inverse modelling of the bromide breakthrough curves.

* Mean K_{sat} is the mean saturated hydraulic conductivity for a given horizon.

** The values of the hydrodynamic parameters obtained from inverse modelling result from adjustments of a dual permeability model except for the column H2-H for which a dual porosity + mobile-immobile model was used. θ_{r_f} always equals 0. For H2-H the saturated water content of immobile compartment: $\theta_{s_{IM}} = 0.35$, the K_{s_s} and K_{s_f} are the K_{sat} values in the slow and fast porous compartments and the K_{sat} of the immobile compartment is null.

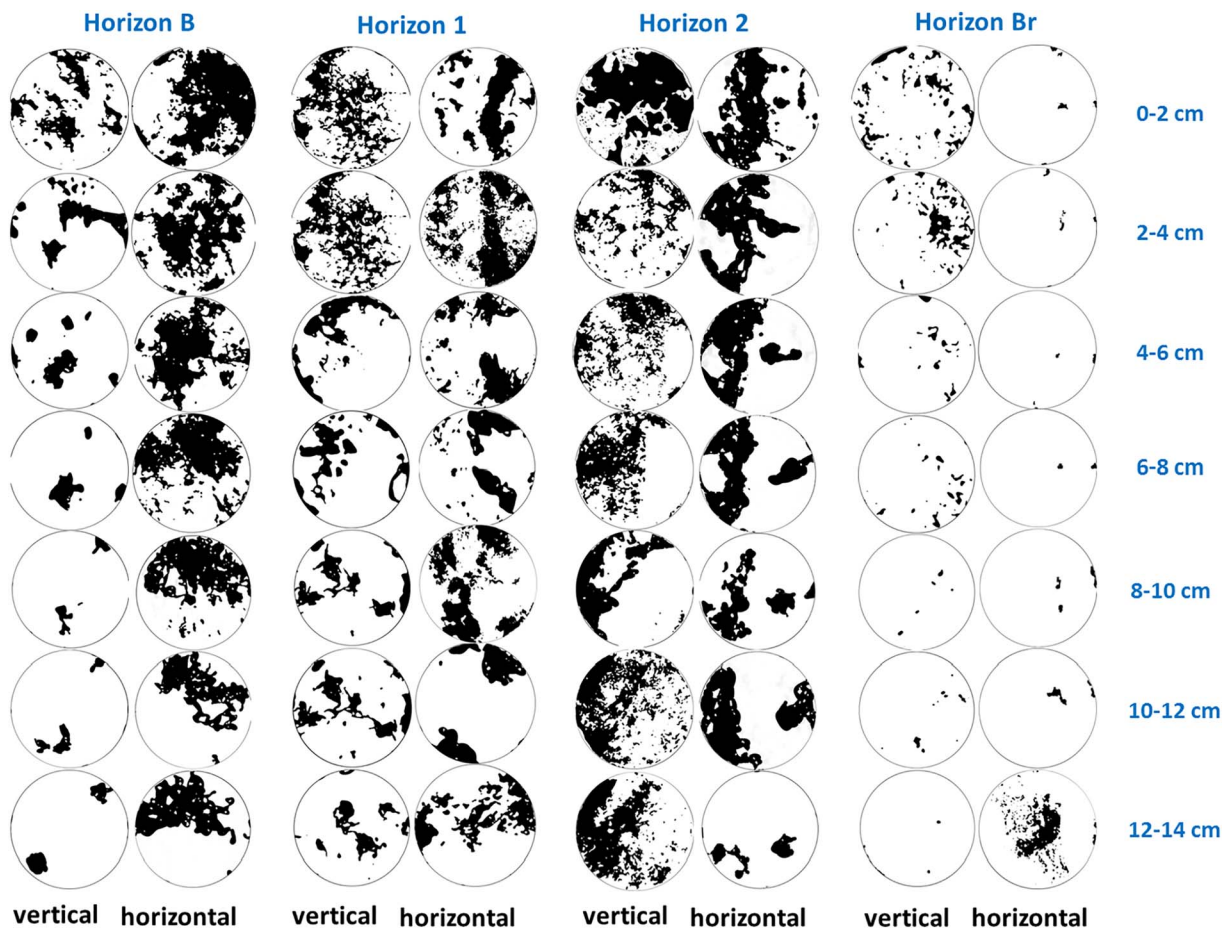


Fig. 5. Imaging of preferential flow patterns within the soil columns. The black areas represent the stained areas at different depths along the soil cores.

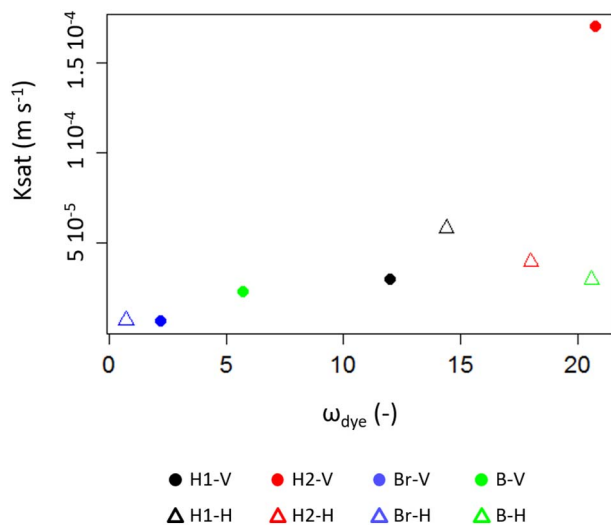


Fig. 6. Evolution of the hydraulic conductivity with the active macroporosity fraction in the set of soil columns.

of the water flow and the bromide leaching pattern (Table 4), the estimated ω were statistically equivalent to the one obtained with dye tracing (linear correlation: slope = 0.97, intercept = 0, $R^2 = 0.92$, p -value = $5 \cdot 10^{-5}$). This highlights the reliability of the simulated parameters, despite the equifinality issue inherent to the large number of fitted parameters.

For all displacement experiments, quantifiable bromide concentrations were measured in the first 50 ml of leachates, which were

collected between 15 s and 7 min after injection began. This suggests that preferential flow occurred in all of the columns (e.g., Paradelo et al., 2016), which is consistent with the observation of connected macroporosity in the columns. Additionally, two major shapes of breakthrough curves can be distinguished.

In the columns of horizons B and H, the curve features a gentle increase and decrease in concentration during the injection and rinsing phases, respectively, as well as a maximum concentration that is less than the injected concentration. If we assume that macropore flow occurred almost instantaneously at a concentration close to the injected concentration, it follows that, throughout the displacement experiment in these columns, matrix flow was a significant contributor to outflow, as bromide concentration remained below the injected concentration. Although the volume of the injected bromide solution was chosen to be larger than the overall pore volume of the columns, this volume was likely not sufficient to ensure a renewal of matrix pore water. This hypothesis is confirmed by the modelling results indicating that even if preferential flow contributed up to 77% to the global outflow for this group of columns, the hydraulic conductivity of the fracture never exceeded 25 times that of the matrix (Table 4).

The other breakthrough curve shape is observed in the columns of the Br horizon and exhibits a sharp increase and early plateau in bromide concentrations during the injection phase, in which the plateau concentration is close to the injected concentration value. Additionally, a sharp decrease in concentration is observed during the rinsing phase. This pattern is not consistent with the small observed macroporosity of the Br horizon but can be explained by the very poor permeability of the soil matrix. In this case, most of the flow bypasses the soil matrix and flows through a few connected macropores (Köhne et al., 2006; Vanderborght et al., 2002). This hypothesis is confirmed by the

modelling results indicating that preferential flow contributed to > 85% to the global outflow for this group of columns and that the hydraulic conductivity of the fracture was > 90 time higher than that of the matrix (Table 4).

In accordance with recent studies relating soil macroporosity and hydraulic conductivity in structured soils (e.g., Katuwal et al., 2015; Paradelo et al., 2016; Udawatta and Anderson, 2008), K_{sat} generally rises along with an ω increase (Fig. 6). As ω is related to the root channels density, which decreases with the distance from the ditch surface, the saturated hydraulic conductivity is overall greater in the upper horizons of the ditch bed than in the banks. The contribution of preferential flow to the global outflow, is however greater in the deep bed and bank horizon. This can be explained by the contrasted hydraulic conductivity of the macroporal compartment relative to that of the matrix compartment (K_{sf}/K_{s_g}) and by ω ($R^2 = 0.95$, p -value = $6 \cdot 10^{-4}$).

In sum, it is mainly in their upper horizons that the bank and ditch bed profiles differ in their patterns of water and solute transport. The top horizons of the ditch bed exhibit larger transport properties due to their larger active macroporosities, which are related to their denser rooting patterns. Thus, in contrast with the soil profile from which it originates, the ditch bed profile exhibits larger infiltration and percolation capacities. However, the deeper percolation of water and solutes is limited in both profiles by their common bottom Br horizon, which exhibits low permeability. The differences in the flow patterns may induce significant contrasts in the transfer and retention of herbicides. Indeed, water pathways determine the material surface area that is in contact with the soil solution and its effective contact time during downward seepage. This conditions the herbicide sorption equilibria.

3.3. Pesticide sorption heterogeneities among the ditch bed and bank soil profiles

The heterogeneities in herbicide sorption properties among the horizons are presented in Table 5. The H1 and H2 horizons exhibit the greatest diuron adsorption capacities and lowest desorption capacities, whereas adsorption on B and Br is low and very easily reversible. Therefore, the sorption capacities of the ditch bed profile are larger than those of the bank profile. For glyphosate, the adsorption coefficient of the B horizon is higher than that of the H1 horizon but is lower than that of the H2 horizon, and the desorption hysteresis of the B horizon is smaller and larger than those of the H1 and H2 horizons, respectively. The Br horizon exhibits a lower adsorption coefficient than the B horizon, but they both exhibit a similar desorption hysteresis. Based on the properties of these horizons, it remains unclear whether the ditch bed profile or the bank profile has the greater retention capacity.

The sorption of herbicides on soils depends on the physicochemical properties of both the molecules and the soils (e.g., Wauchope et al., 2002; Weber et al., 2004). Differences in the physicochemical properties among the soil horizons of the 2D cross-section of the ditch (see

Section 3.1) can thus result in different degrees of herbicide sorption between the ditch bed and bank materials. Organic matter in soils features preferential sorption sites for hydrophobic molecules, such as diuron (Dollinger et al., 2016; Weber et al., 2004). Therefore, the decrease in the diuron adsorption coefficients with depth along the 2 profiles is consistent with decreasing organic carbon content (Tables 1 and 5). This decrease is more progressive, and less marked, in the bank profile than in the ditch bed profile. Moreover, although few studies have reported that organic matter contributes to (e.g., Gros et al., 2017; Okada et al., 2016) or limits (e.g., Ololade et al., 2014) the sorption of the hydrophilic herbicide glyphosate, meta-analyses of the relations between glyphosate sorption coefficients and soil properties over a wide range of soils (Dollinger et al., 2015b; Vereecken, 2005) and a range of ditch soils (Dollinger et al., 2015b) suggest that glyphosate sorbs mainly on the mineral fraction of soils by forming complexes with soil cations. This explains why it doesn't follow the same sorption gradients as diuron in both profiles.

In summary, the heterogeneities in the sorption coefficients of the two studied pesticide within a given profile are more substantial under the ditch bed than in the banks. Generally, due to the enrichment in organic matter of the ditch bed horizons, the sorption capacities of hydrophobic molecules in the ditch bed profile should be greater than those in the bank profile. Concerning ionisable compounds with low hydrophobicity, a higher sorption capacity of ditch bed profiles is not straightforward. The desorption hysteresis are generally significant in the ditch bed and are weaker or null in the bank soils. This should lower the release of pesticides previously adsorbed in the ditch bed soils as compared to the bank soils.

4. Conclusions

Ditch networks greatly influence the hydrological processes and contaminant dynamics within cropped catchments. Indeed, ditch soils regulate most of the surface to subsurface water and pesticide exchanges. This study provides the first description of the range of soil properties influencing the magnitude of the water and pesticide exchanges occurring between surface water and groundwater along a ditch cross section profile. These ditch bed soil properties were also for the first time compared with those of the surrounding field soils. The in-situ and laboratory characterization of the physico-chemical properties evidenced distinct soil profiles between both the ditch bed and banks profiles. The ditch bank profile was equivalent to the surrounding field profile. In particular, the ditch bed upper horizons contain up to 2 times more organic carbon than the bank soils. These upper ditch bed horizons being also located closer to the ditch surface than the bank soils, they contain a denser network of plant roots which increases their active macroporosity and in turn their hydraulic conductivity. The deeper horizons share however, great similarities in both profiles, in particular their poor macroporosity, hydraulic conductivity and organic carbon content.

In conclusion, the physicochemical and sorptive properties of specific ditch bed horizons contrast with those of the ditch banks and neighbouring field soils. These differences may thus have different effects on the risk of groundwater contamination by pesticides. On one hand, ditch beds exhibit higher organic matter contents than field soils, possibly limiting the percolation of hydrophobic pesticides due to increased retention in the soil matrix. On the other hand, the upper horizons of ditch beds present larger active macroporosity and transport property values, which favour percolation. The final balance between the two effects, in terms of overall groundwater contamination risk, depends on the local hydrological conditions. Nevertheless, this case study highlights the need to investigate the specific properties of the ditch beds, when studying and modelling water and pesticide fates in cropped landscapes with ditches. Especially, extrapolating to the ditch materials the soil properties determined from field soil maps or from measurements in nearby field soils is not enough.

Table 5
Sorption coefficients of the herbicides on ditch soils.

Molecule	Horizon	Kf_{ads} $\mu g^{(1-n)}l^n$ kg^{-1}	n_{ads}	Kd_{ads} $l\ kg^{-1}$	Kf_{des} $\mu g^{(1-n)}l^n$ kg^{-1}	n_{des}	H
Diuron	B	4.49	0.84	1.86	110.58	0.12	0.98
	H1	14.79	0.83	5.17	114.66	0.32	0.39
	H2	10.57	0.82	3.52	118.74	0.25	0.31
	Br	4.12	0.81	1.49	114.31	0.09	0.97
Glyphosate	B	157.66	0.94	109.90	666.99	0.28	0.29
	H1	77.60	0.93	51.69	675.64	0.14	0.15
	H2	165.24	0.96	129.86	533.29	0.36	0.37
	Br	124.36	0.91	75.38	505.00	0.29	0.32

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